A FINITE ETALE SCHEME COVER YIELDS A DENSE OPEN SUBSPACE WHICH IS A SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2436-2447, immediately following the statement at 2428-2434.

Editorial note. The First proof of this lemma is counted; the author's Second proof (source lines 2449 onward) is a separate independent argument and is not part of this card.

Editorial note. One same-chapter proposition pointer is recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Finite etale is finite locally free, with degree decomposition. standard fact used at proof lines 2437-2441: U -> X finite etale decomposes X into open and closed X_d where the cover has constant degree d. Reduces to constant degree d, which feeds the universally-bounded fibres condition.
- Reasonable spaces have dense open scheme subspace. Proposition proposition-reasonable-open-dense-scheme of this chapter, cited at proof lines 2445-2447. The terminal input: reasonableness of X gives the dense open scheme.

Source reference. Original proposition 03JI: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2273–2273. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. If there exists a finite, étale, surjective morphism $U \to X$ where U is a scheme, then there exists a dense open subspace of X which is a scheme.*

First proof. The morphism $U \to X$ is finite locally free. Hence there is a decomposition of X into open and closed subspaces $X_d \subset X$ such that $U \times_X X_d \to X_d$ is finite locally free of degree d. Thus we may assume $U \to X$ is finite locally free of degree d. In this case, let $U_i \subset U$, $i \in I$ be the set of affine opens. For each i the morphism $U_i \to X$ is étale and has universally bounded fibres (namely, bounded by d). In other words, X is reasonable and the result follows from Proposition 03JI. □

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.